MySQL Assignment 2

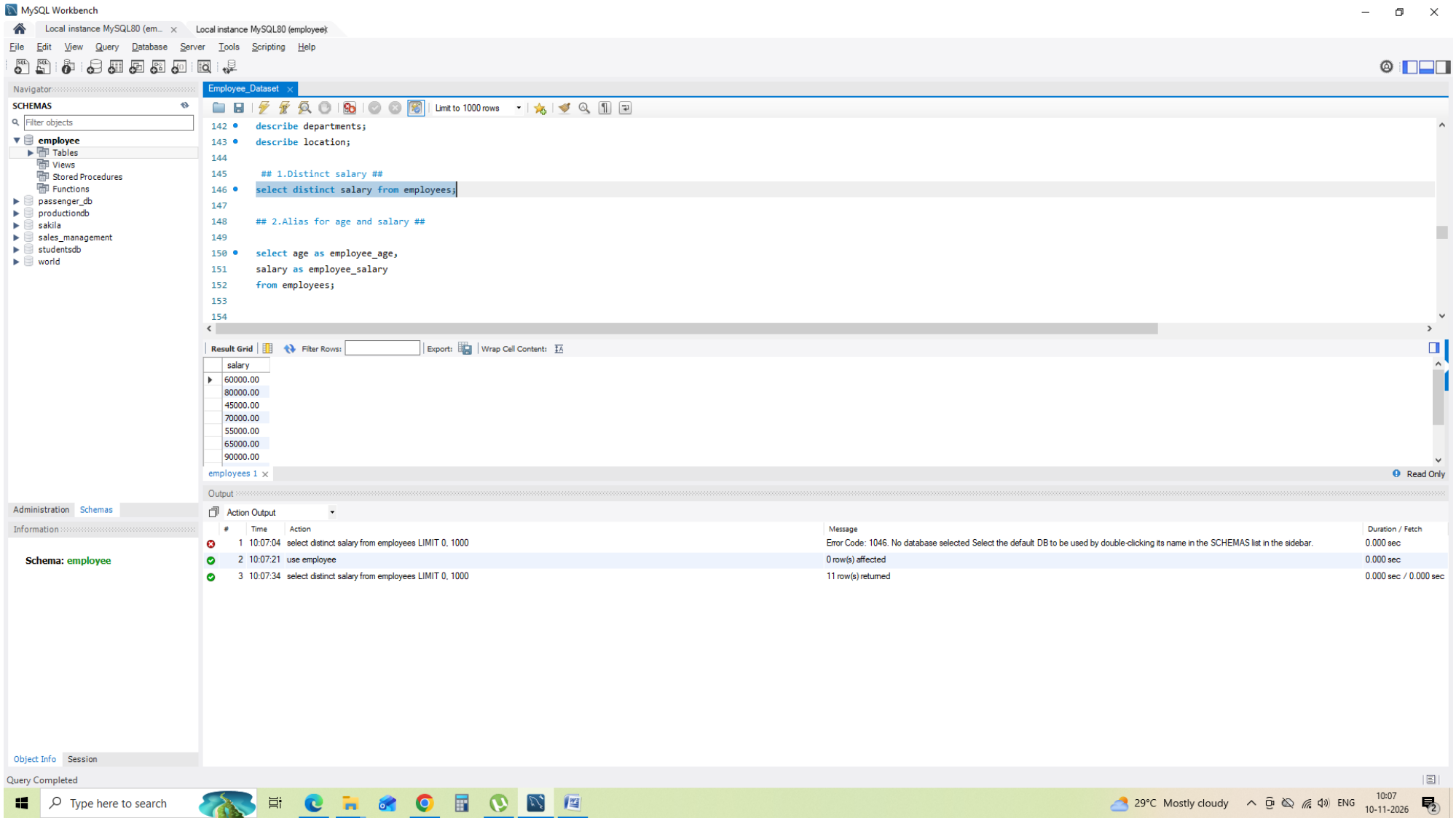
Querying Data – Screenshots with Headings and Explanations

Database: employee | Tool: MySQL Workbench

The supplied screenshots are arranged in document order. Each page includes a related heading and a short explanation of the SQL operation or result shown.

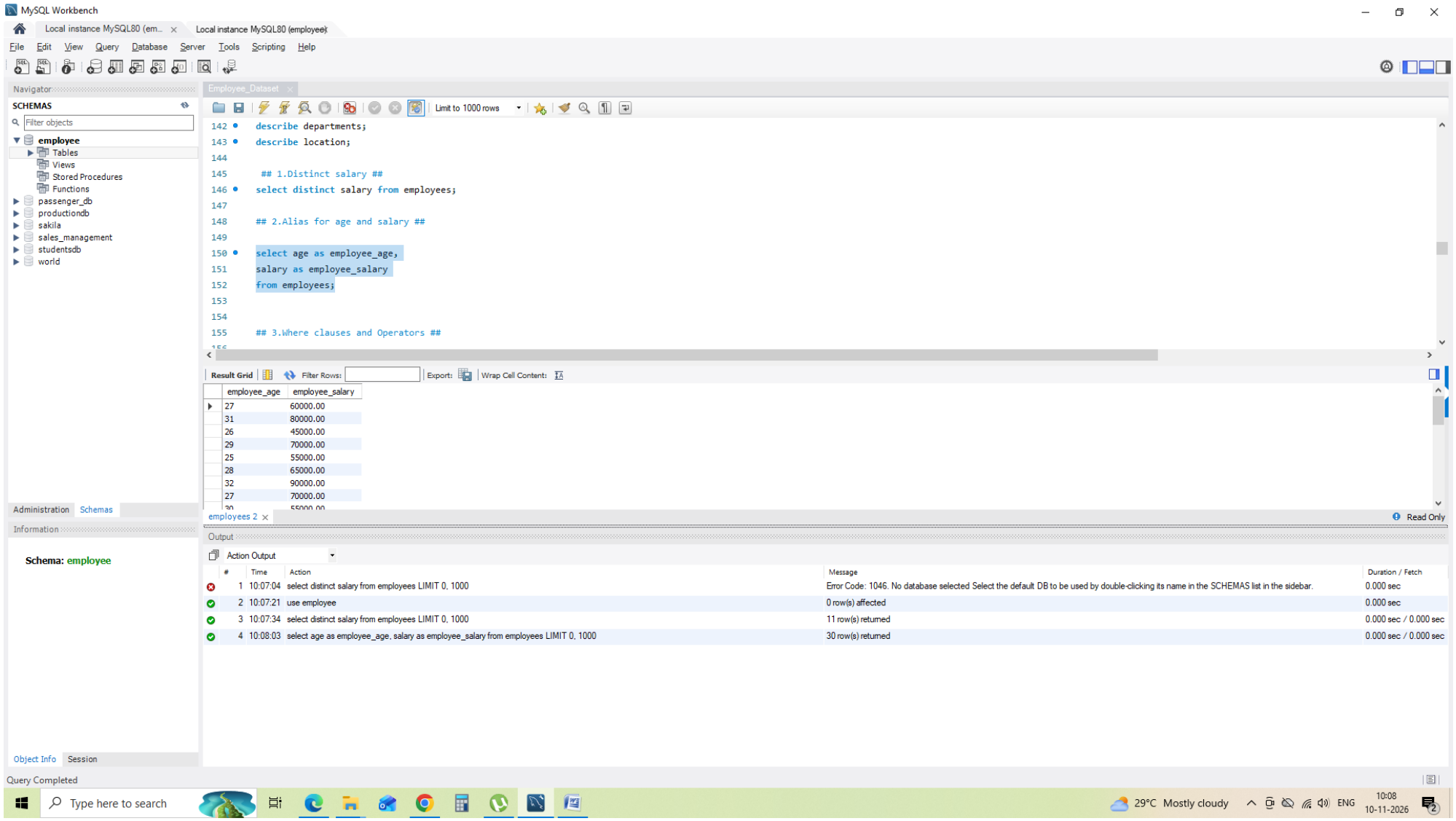
1. Distinct salaries and column aliases

DISTINCT returns each salary value once. AS gives the age and salary columns the output headings Employee_Age and Employee_Salary.



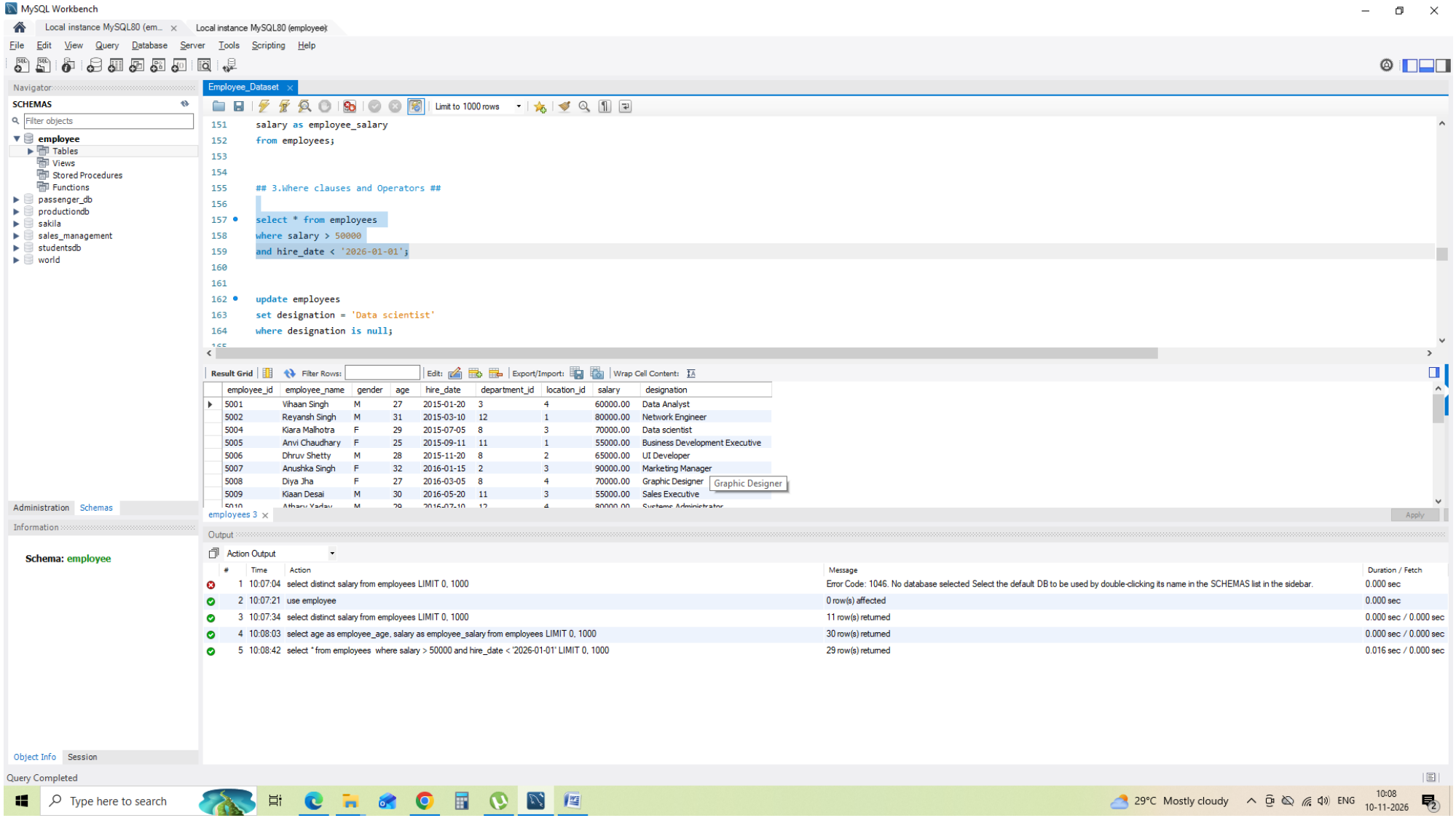
2. Filtering employees with WHERE

WHERE filters employee rows using both conditions: salary greater than ■50,000 and a hire date before the stated cutoff.



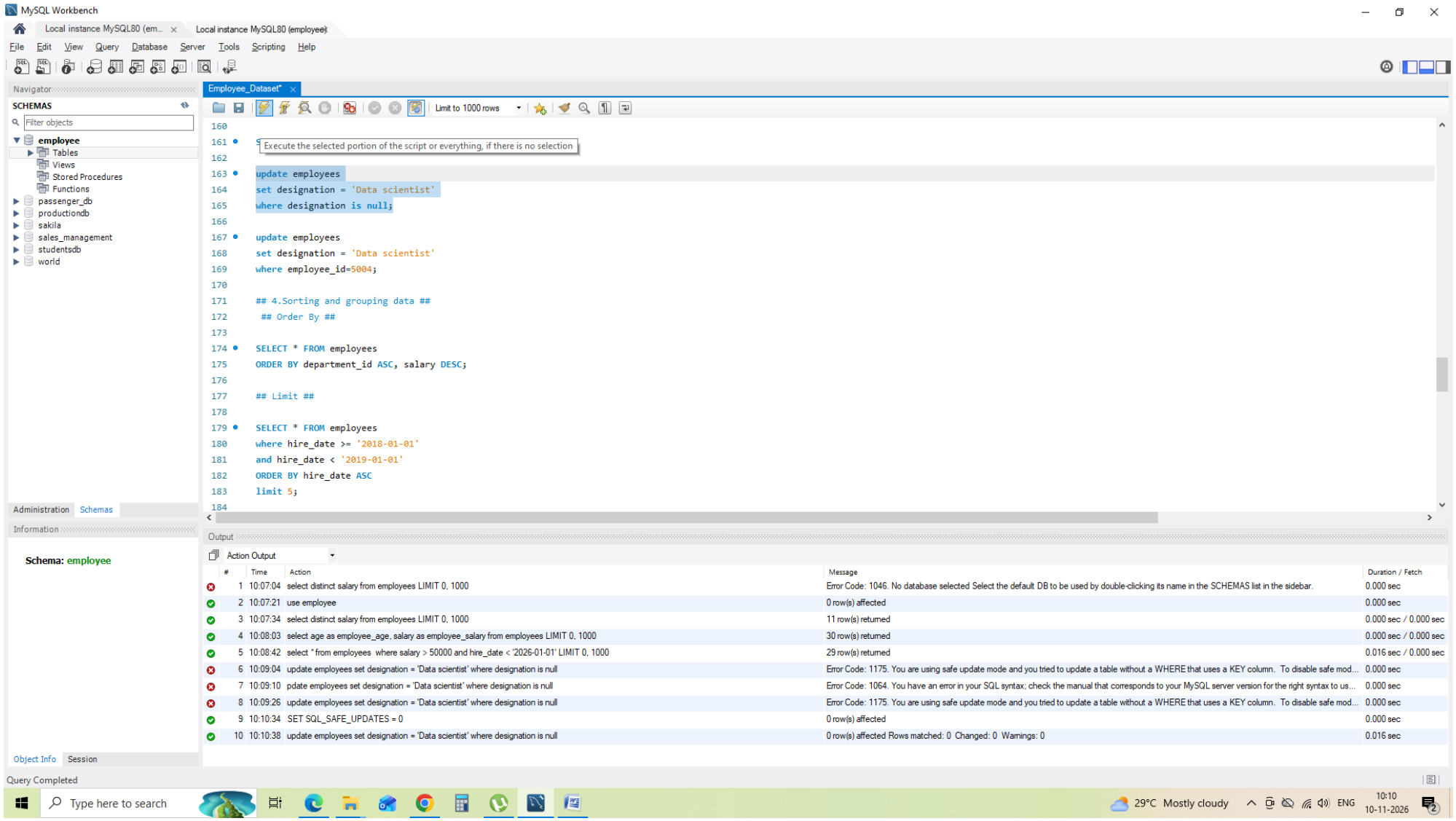
3. Updating a missing designation

UPDATE fills NULL designation values with “Data Scientist”. The screenshot also shows Safe Update Mode handling. Verify the affected rows and use the correct employee key if updating one specific employee.



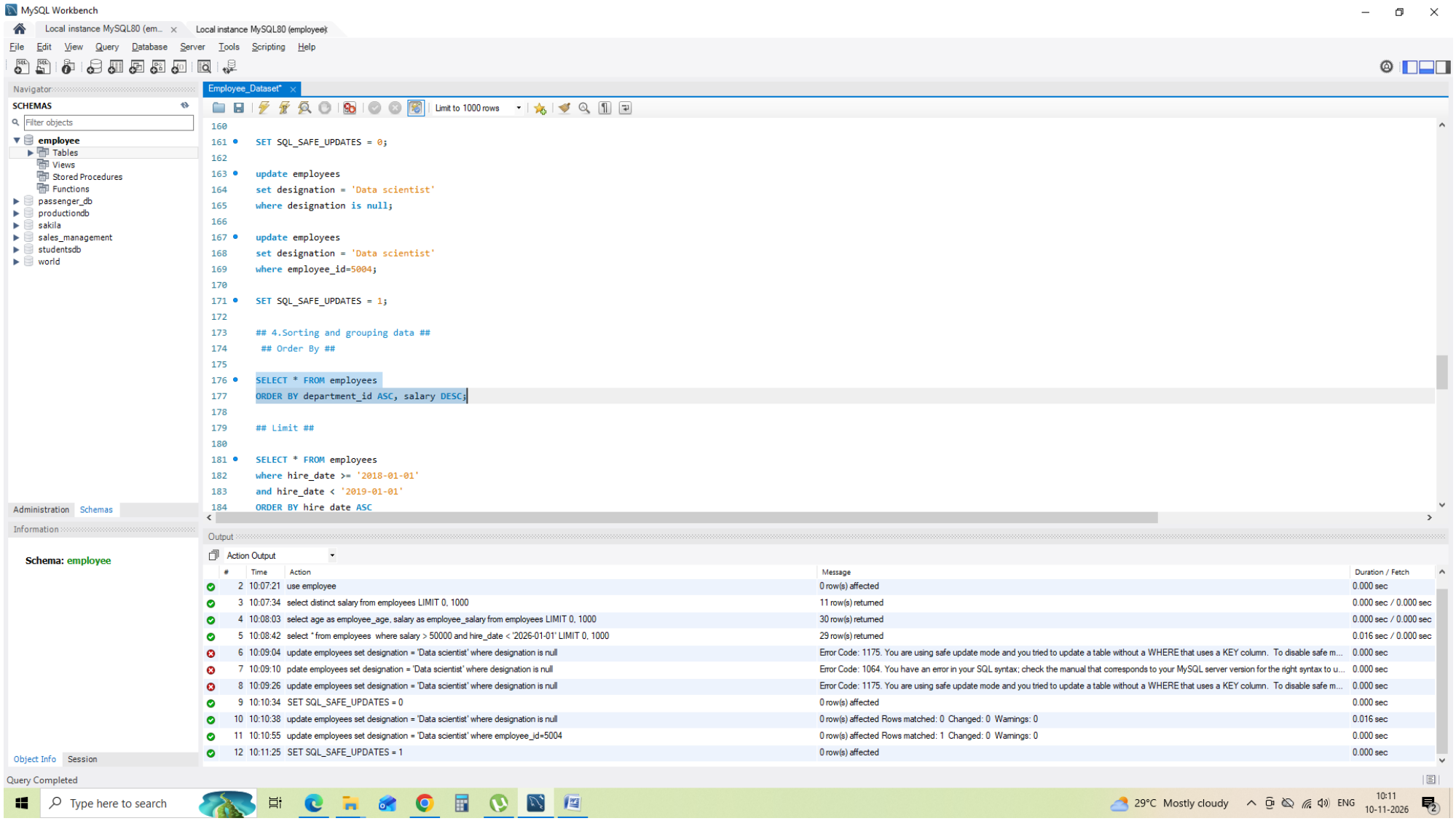
4. Sorting and selecting the first five 2018 hires

ORDER BY sorts department_id ascending and salary descending. The LIMIT query filters hires within 2018, sorts by hire date, and returns up to five rows.



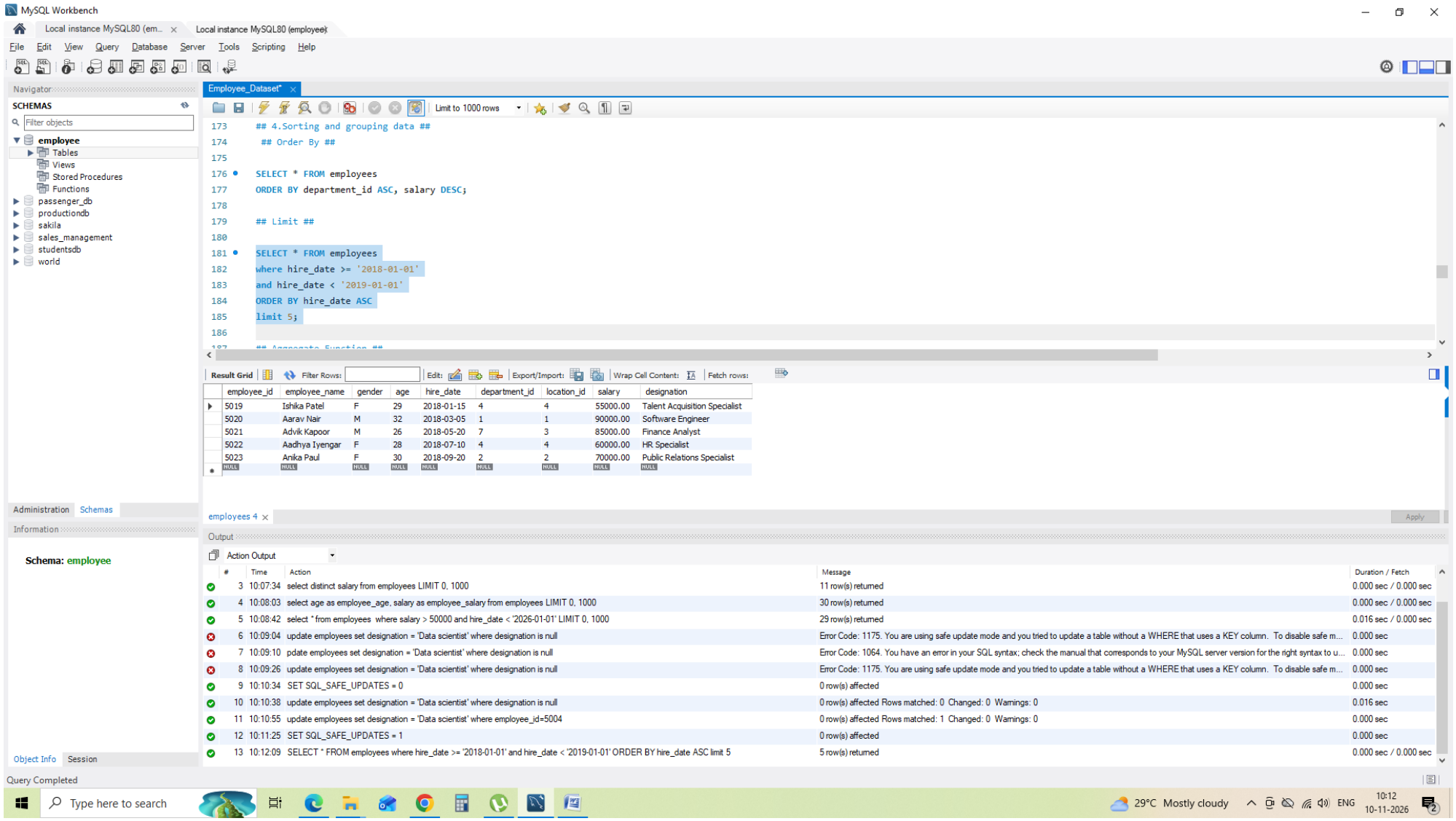
5. Aggregate functions: Finance salary and minimum age

SUM() calculates total salary for Finance employees by joining Employees and Departments. MIN() returns the smallest age in Employees.



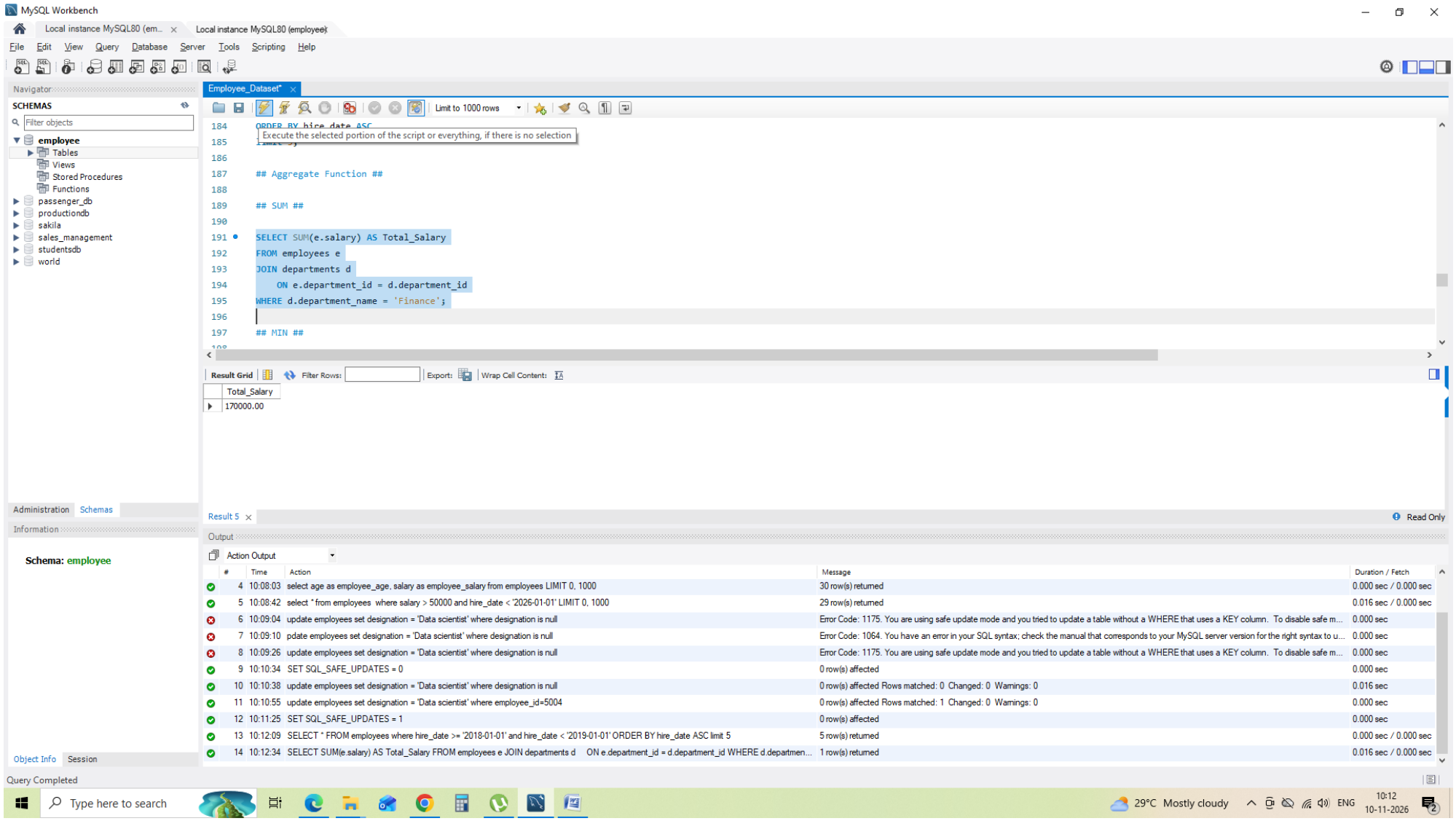
6. GROUP BY: maximum salary by location

The query groups employees by location and uses MAX(salary) to return the highest salary at each location.



7. GROUP BY: average salary for Analyst designations

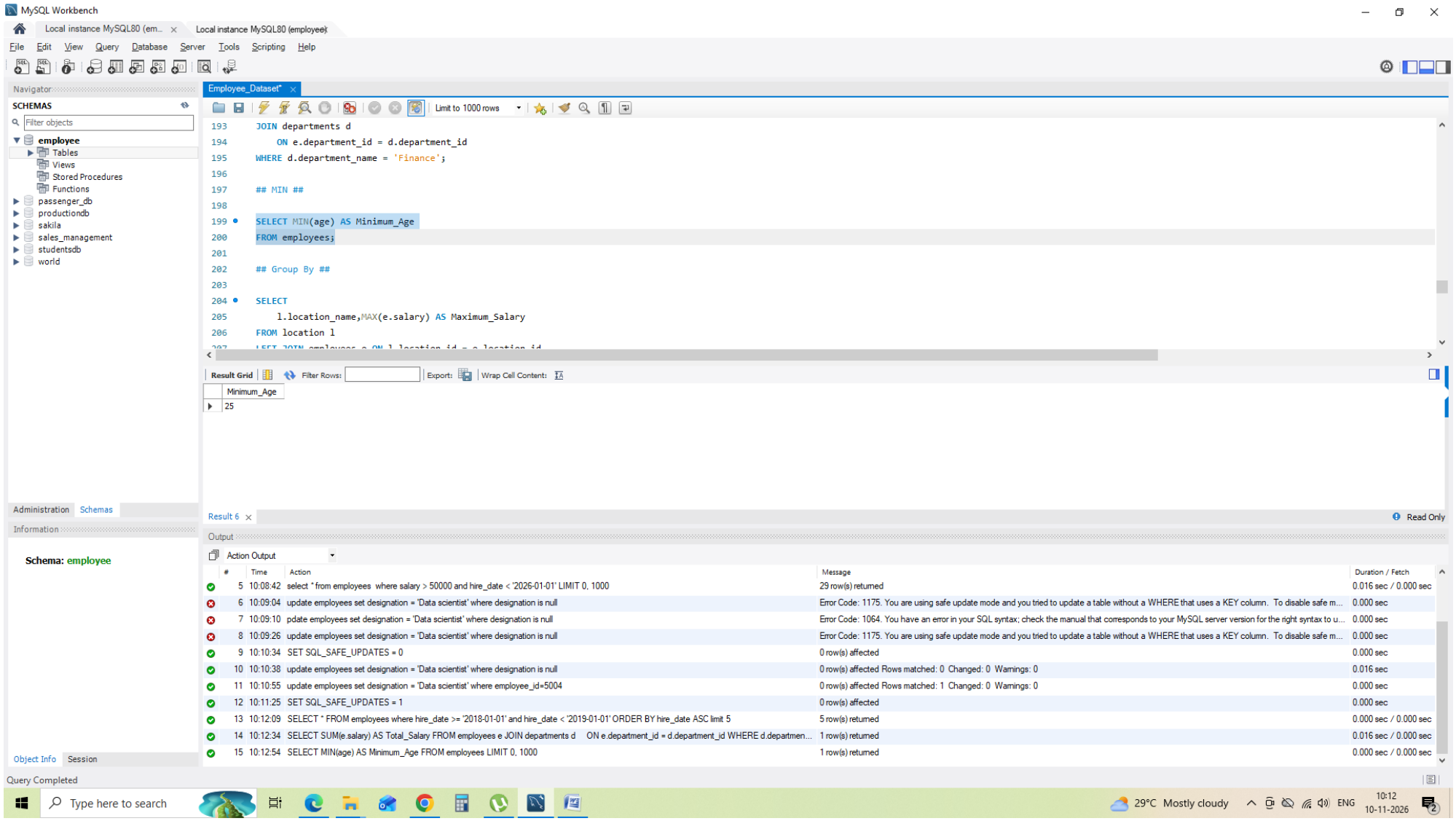
LIKE '%Analyst%' selects designations containing “Analyst”. GROUP BY calculates a separate average salary for each matching designation.



#	Time	Action	Message	Duration / Fetch
4	10:08:03	select age as employee_age, salary as employee_salary from employees LIMIT 0, 1000	30 row(s) returned	0.000 sec / 0.000 sec
5	10:08:42	select * from employees where salary > 50000 and hire_date < '2026-01-01' LIMIT 0, 1000	29 row(s) returned	0.016 sec / 0.000 sec
6	10:09:04	update employees set designation = 'Data scientist' where designation is null	Error Code: 1175. You are using safe update mode and you tried to update a table without a WHERE that uses a KEY column. To disable safe m...	0.000 sec
7	10:09:10	pdate employees set designation = 'Data scientist' where designation is null	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to u...	0.000 sec
8	10:09:26	update employees set designation = 'Data scientist' where designation is null	Error Code: 1175. You are using safe update mode and you tried to update a table without a WHERE that uses a KEY column. To disable safe m...	0.000 sec
9	10:10:34	SET SQL_SAFE_UPDATES = 0	0 row(s) affected	0.000 sec
10	10:10:38	update employees set designation = 'Data scientist' where designation is null	0 row(s) affected Rows matched: 0 Changed: 0 Warnings: 0	0.016 sec
11	10:10:55	update employees set designation = 'Data scientist' where employee_id=5004	0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0	0.000 sec
12	10:11:25	SET SQL_SAFE_UPDATES = 1	0 row(s) affected	0.000 sec
13	10:12:09	SELECT * FROM employees where hire_date >= '2018-01-01' and hire_date < '2019-01-01' ORDER BY hire_date ASC limit 5	5 row(s) returned	0.000 sec / 0.000 sec
14	10:12:34	SELECT SUM(e.salary) AS Total_Salary FROM employees e JOIN departments d ON e.department_id = d.department_id WHERE d.departmen...	1 row(s) returned	0.016 sec / 0.000 sec

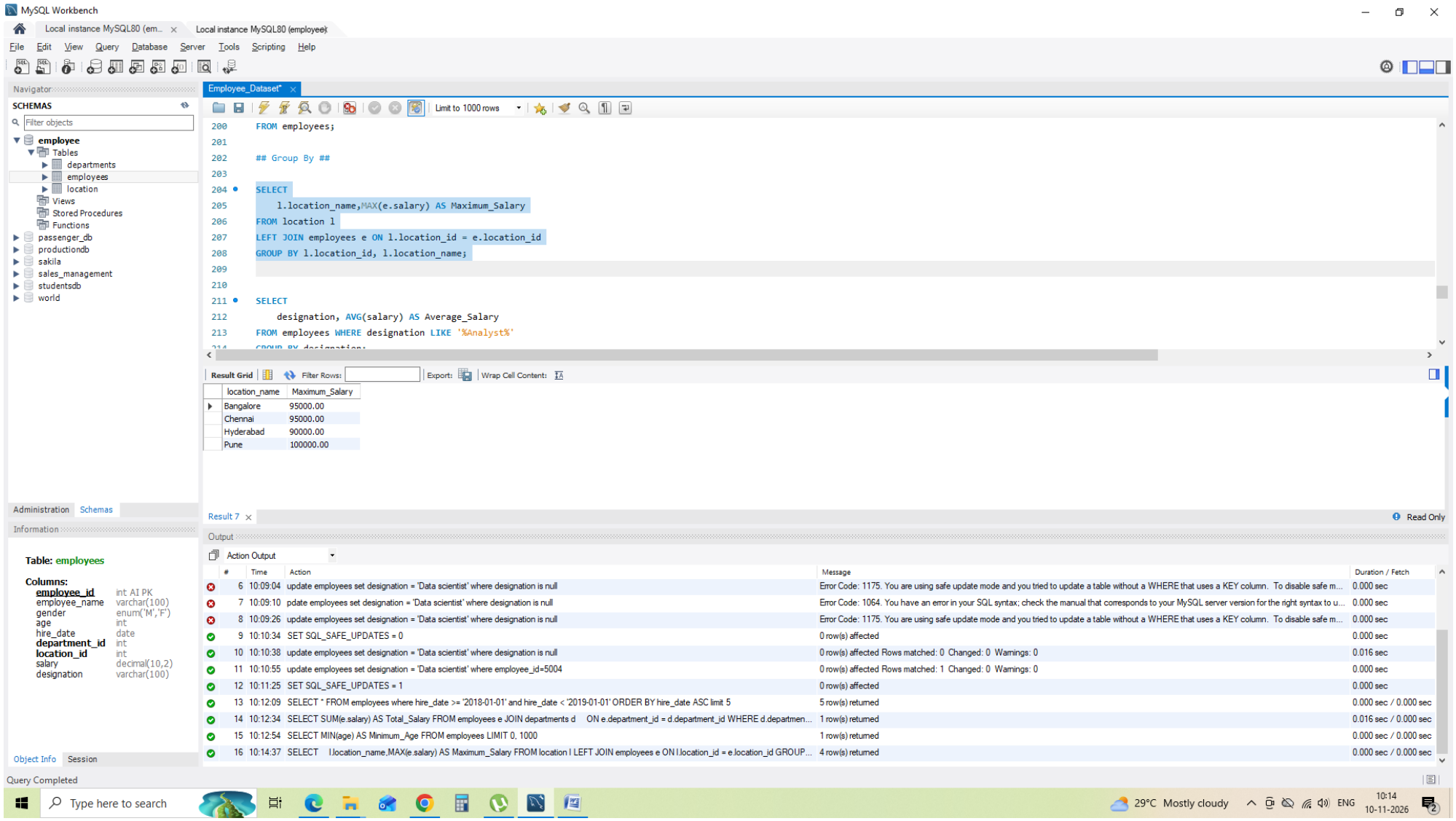
8. HAVING: departments with fewer than three employees

LEFT JOIN keeps every department, including those with no employees. COUNT() counts matched employee IDs, and HAVING filters groups with fewer than three.



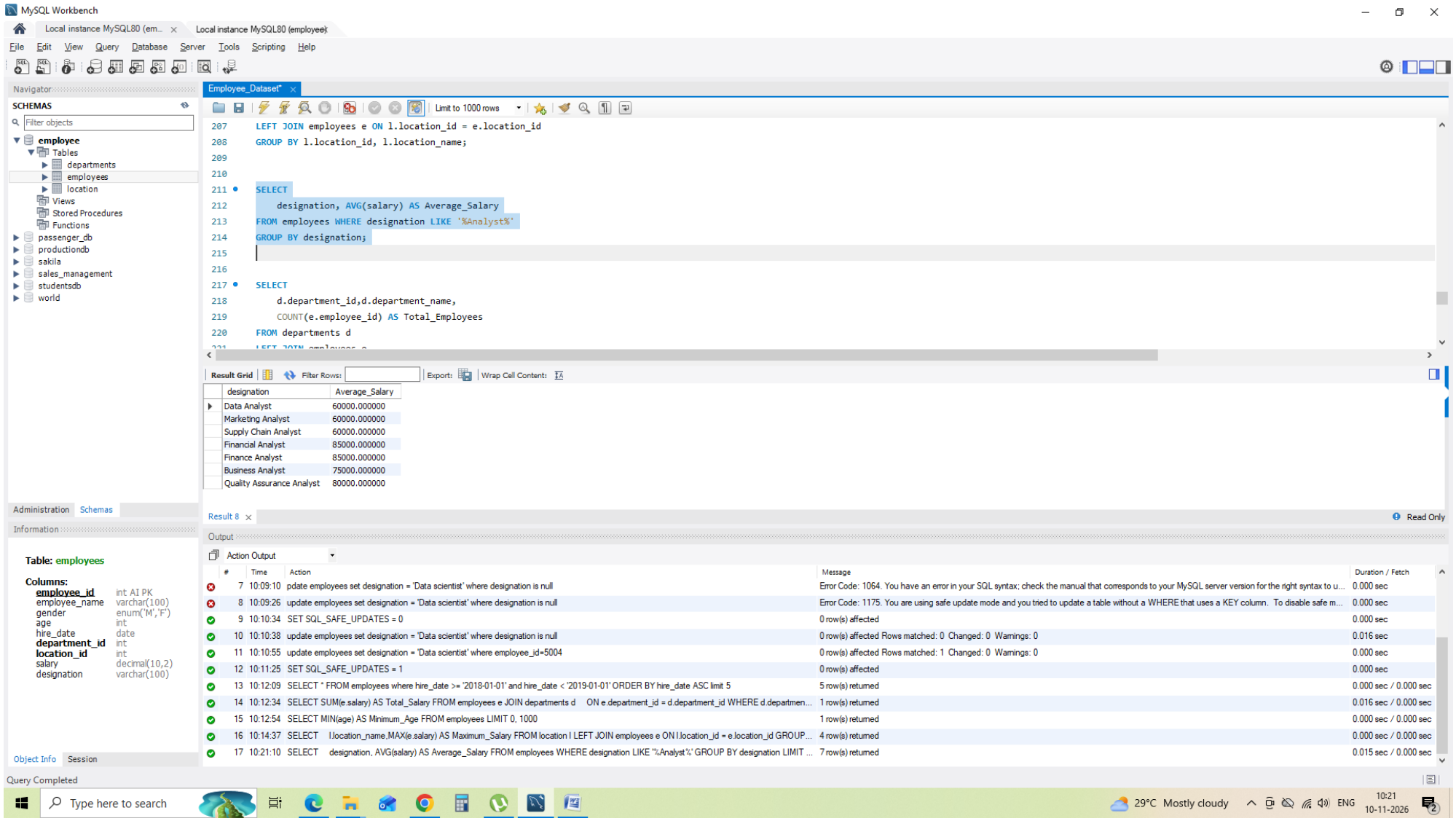
9. HAVING: locations with female employees under age 30 on average

The query filters female employees, groups them by location, and returns locations where AVG(age) is below 30.



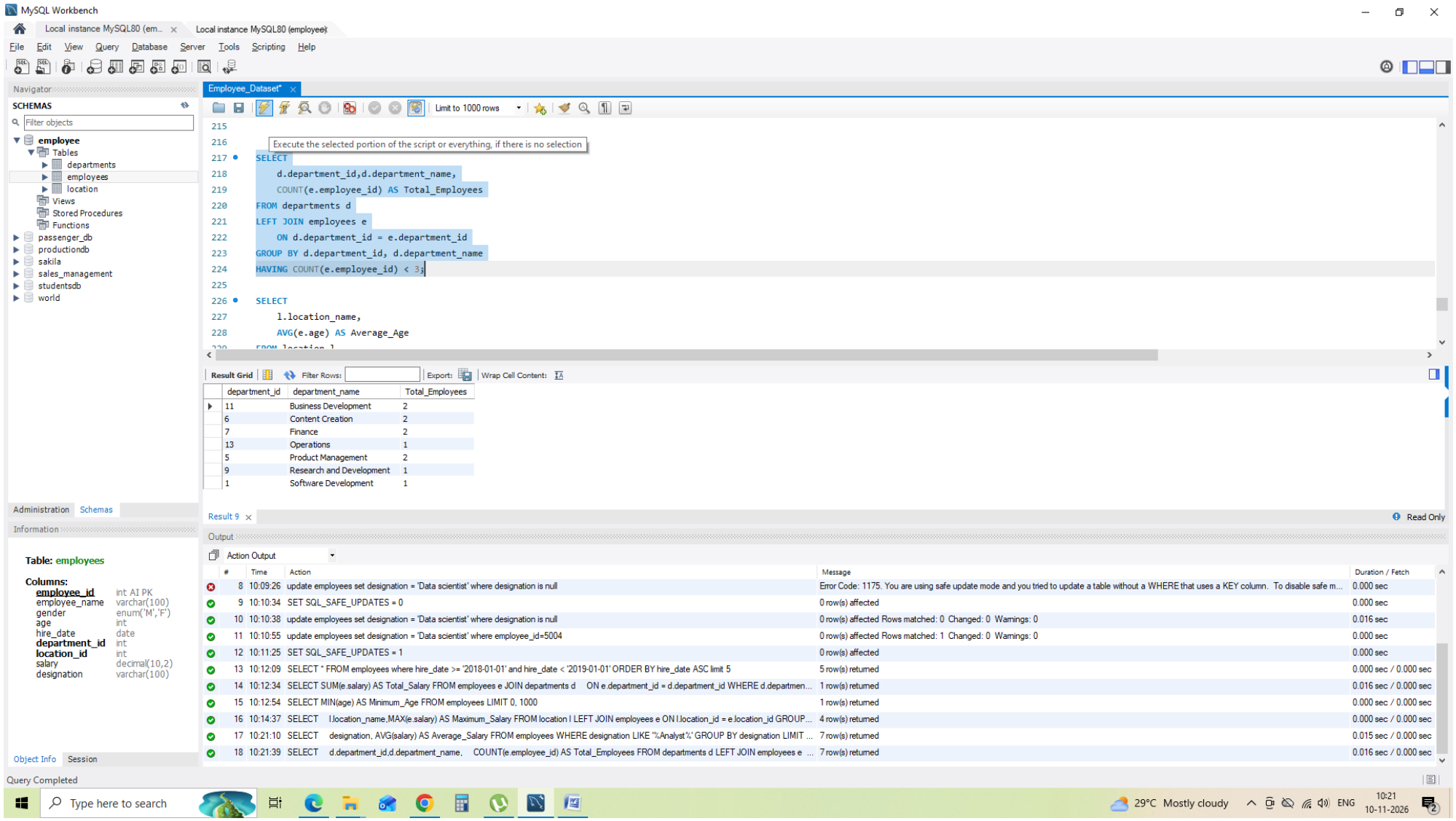
10. INNER JOIN: employees and department names

INNER JOIN returns employees with a matching department. The output includes employee name, designation, and department name.



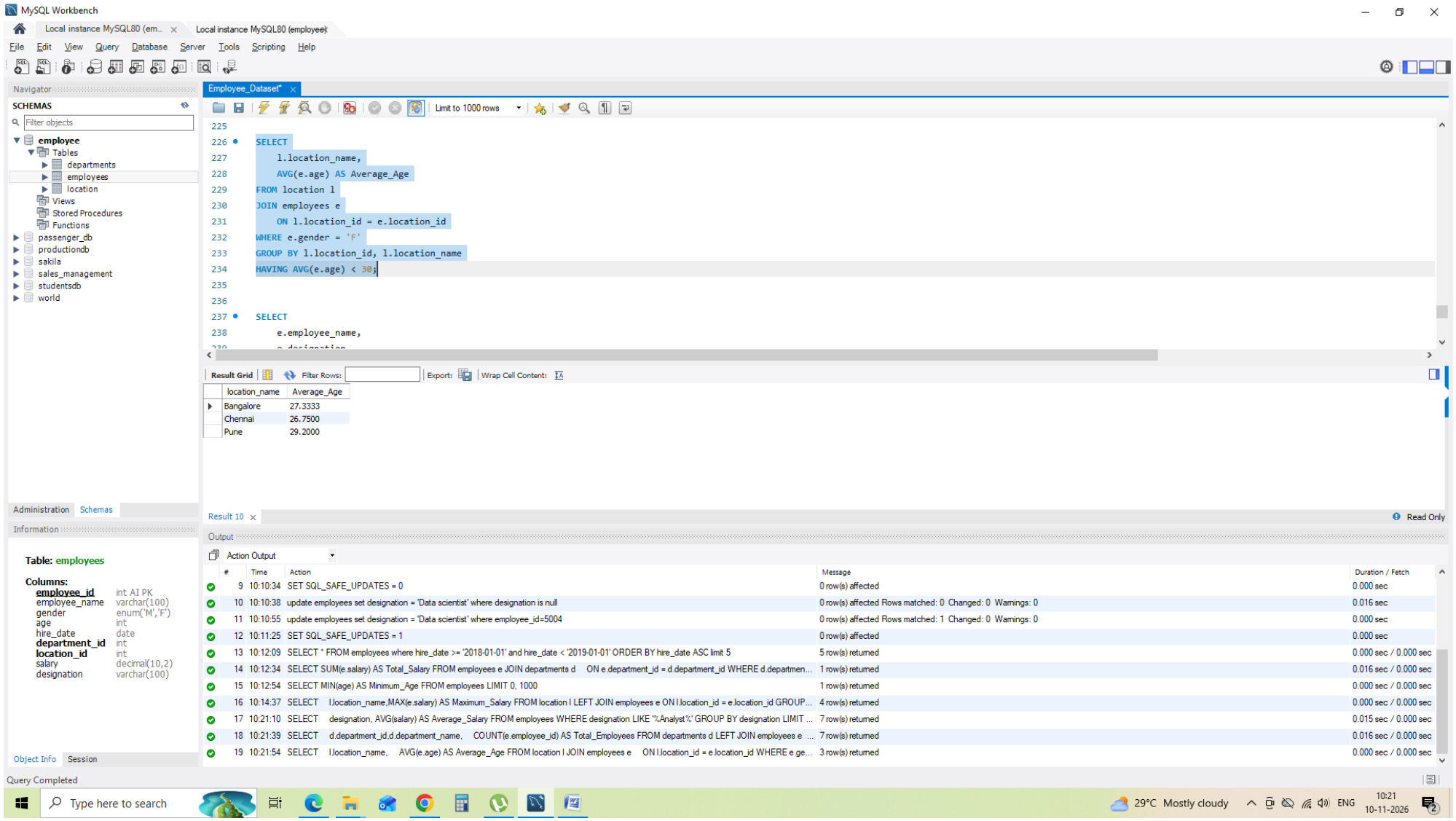
11. LEFT JOIN: employee count for every department

LEFT JOIN preserves all departments. COUNT(employee_id) gives each department's employee count; departments with no employees should show zero.



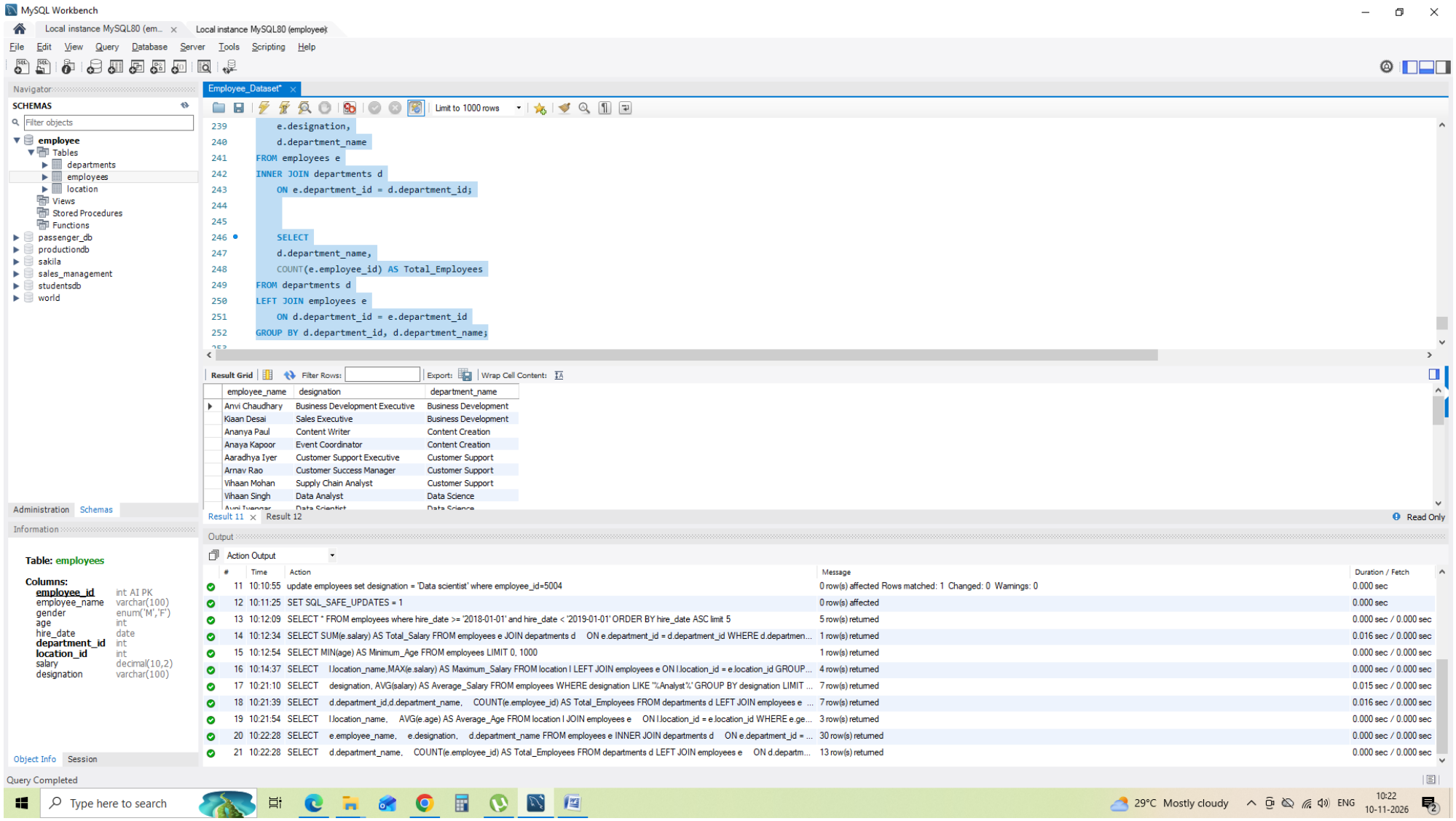
12. RIGHT JOIN: all locations and assigned employees

RIGHT JOIN preserves every location and matches employees by location ID. A location without employees displays NULL for the employee name.



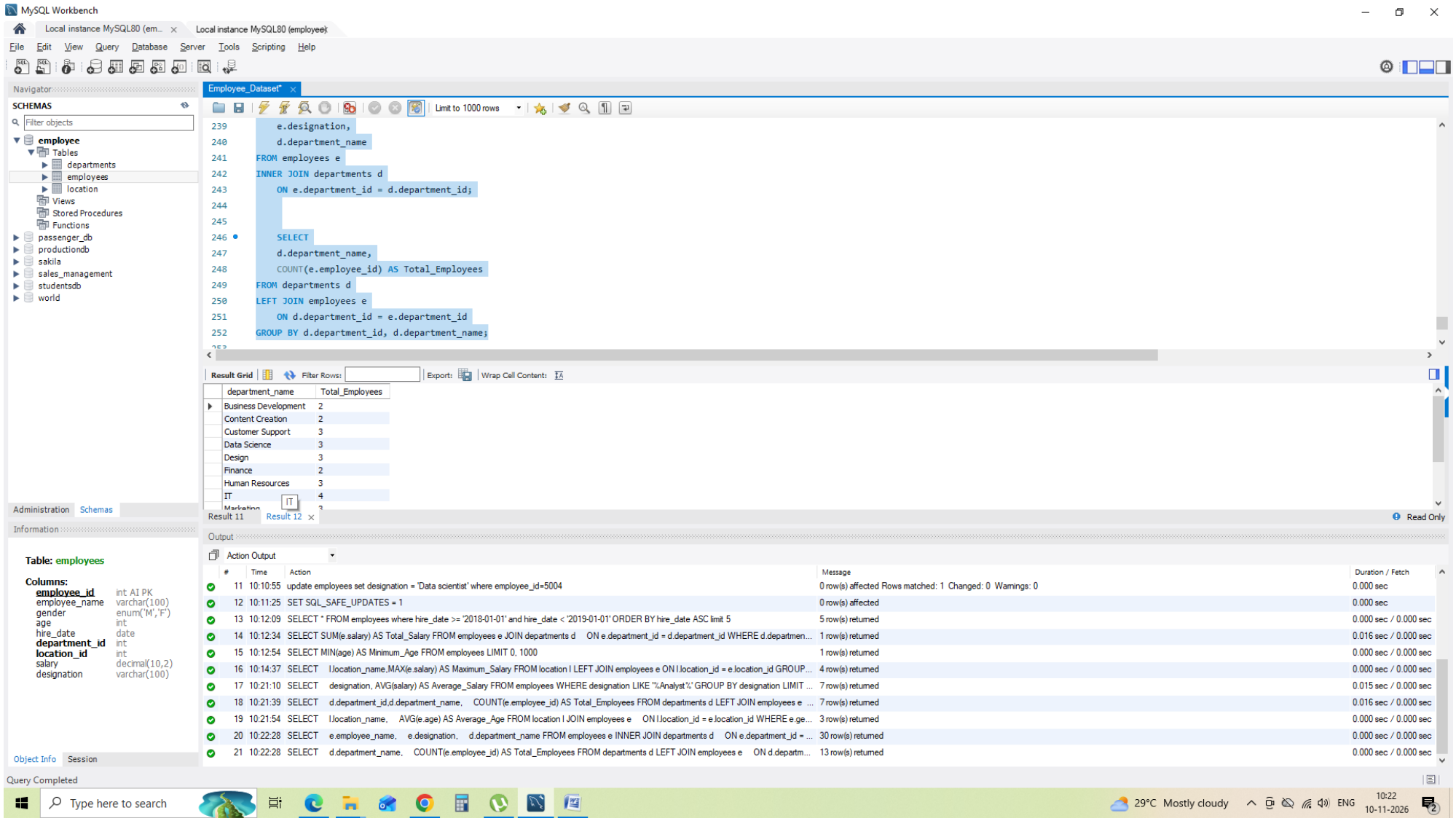
13. Query results: distinct salaries and aliases

The result grids show the output of the DISTINCT salary query and the aliased age and salary query.



14. Query results: sorting, grouping and HAVING

The visible result grids provide output evidence for the hire-date filter, aggregate calculations, location-wise maximum salaries, Analyst averages, and department counts.



15. Query results: joins

The result grids show employee-to-department matches, department-wise employee counts, and locations with their assigned employees.

